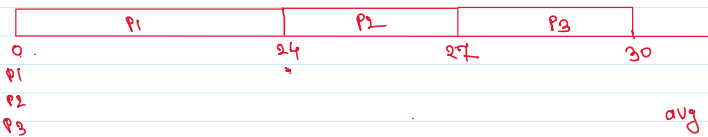


cpu scheduling criteria.

- ① **cpu utilization** - should be max. residential = 60%, server = 90%.
- ② **throughput** - amount of work done per unit time (max).
- ③ **waiting time** - time spend by the process in the ready queue (min).
- ④ **Turn-around time** - arrival to the completion of the process (min)
(cpu time + cpu wait time + io time + io wait).
- ⑤ **response time** - arrival to the 1st cpu time (min).

➤ FCFS Scheduling (non-preemptive algo)

Process	Arrival Time	CPU Burst	Wait Time	Turn Around Time	Response Time
P1	0	24	0	24	0
P2	0	3	24	27	24
P3	0	3	27	30	27

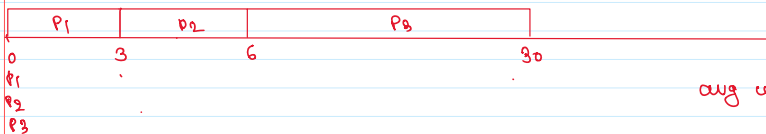


$$\begin{aligned} \text{avg wt time} &= \frac{0 + 24 + 27}{3} \\ &= \frac{51}{3} \\ &= 17 \text{ ms} \end{aligned}$$

Convoy effect - due to the longest process another processes = 17 ms
No + getting enough cpu time.

➤ FCFS Scheduling

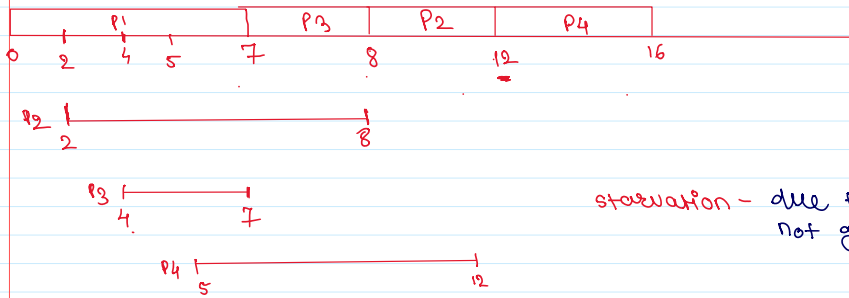
Process	Arrival Time	CPU Burst	Wait Time	Turn Around Time
P1	0	3	0	3
P2	2	3	3	6
P3	4	24	6	30



$$\begin{aligned} \text{avg wt time} &= \frac{0 + 3 + 6}{3} \\ &= \frac{9}{3} = 3 \text{ ms} \end{aligned}$$

➤ SJF/SNTF Scheduling (non preemptive algorithm)

Process	Arrival Time	CPU Burst	Wait Time	Turn Around Time
P1	0	7	0	7
P2	2	4	6	10
P3	4	1	3	4
P4	5	4	7	11



$$\text{avg wt time} = \frac{0+6+8+7}{4}$$

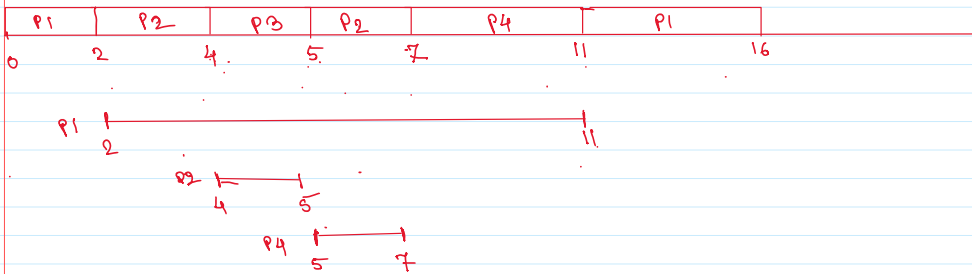
$$= \frac{16}{4}$$

$$= 4 \text{ ms.}$$

starvation - due to shorter process another longer processes not getting enough cpu time.

> SRTF Scheduling (preemptive algorithm).

Process	Arrival Time	CPU Burst	Remaining Time	Wait Time	Turn Around Time
P1	0	7	5	3	16
P2	2	4	2	1	5
P3	4	1	0	0	1
P4	5	4	2	2	6

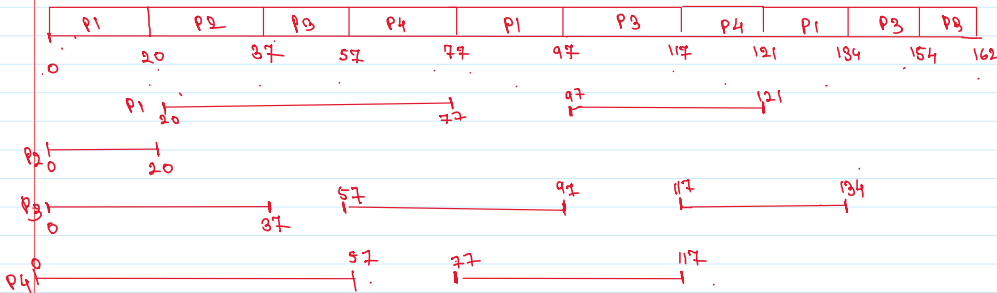


> Round Robin Scheduling (preemptive algo).

time quantum = 20 ms.

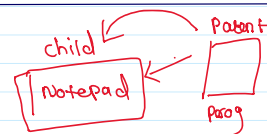
100 ms

Process	AT	CPU Burst	Remaining Time	Wait Time	Response Time
P1	0	53	28 13	57 + 24 = 81	0
P2	0	17	0	20	20
P3	0	68	48 28	37 + 40 = 77	37
P4	0	24	4	57 + 40 = 97	57



> Priority Scheduling (non preemptive algo).

Process	Arrival Time	CPU Burst	Priority	Wait Time
P1	0	10	3	6
P2	0	1	1 highest	0
P3	0	2	4 lowest	16
P4	0	5	2	1



P2	P4	P1	P3
0	1	6	16
			18

starvation:- due to high priority process another process not getting enough cpu time.

Aging :- increase the priority of the process.

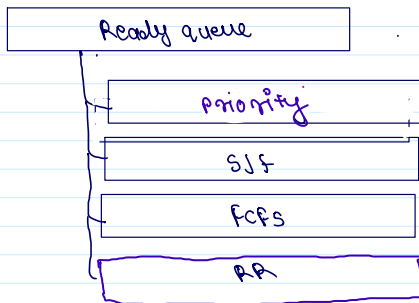
Priority algorithm (preemptive)

Process order

P ₁ (8)	P ₁ (8)
P ₂ (10)	P ₃ (9)
P ₃ (5)	P ₄ (5)
P ₄ (5)	P ₇ (9)
P ₅ (7)	P ₅ (7)
P ₆ (9)	P ₆ (9)
P ₇ (9)	P ₂ (10)

A(5)
B(4)

multilevel queue.



multilevel feedback queue.

